23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I & PHASE – II

Analysis and Architectural Modelling (CO1 & CO2)

Group No : ___________A9_________

Team Members :

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PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

1. Problem Title : __________Course Feedback System _________________________

2. Problem Description : Students provide feedback about courses and instructors at the end of each semester

3. Identified Entities (Classes)

Class Name	Description
Student	Stores details of student
Course	Stores various course’s information
Question	To ask questions regarding course
Feedback	Stores feedback given by students

Admin	To view the feedback reports given by students
FeedbackSystem	It acts as the central controller that stores and manages arraylists and classes.
File Manager	All the operations related to storage and file managing

4. Class Attributes and Behaviours

Class Name	Attributes	Methods (Behaviours)
Student	name : String studentId : String department : String semester : int feedbacks : ArrayList<Feedback>	addFeedback(Feedback f) : void displayStudent() : void viewFeedbacks() : void
Course	courseCode : String courseName : String instructor : String subject : String instructorId : String feedbacks : ArrayList<Feedback>	addFeedback(Feedback f) : void displayCourse() : void viewCourseFeedbacks():void instructorDetails() : void
Question	questionNo : int question : String	displayQuestion() : void
Feedback	student : Student course : Course comment : String ArrayList<Question> questions ArrayList<Integer> ratings	addComment(String comment) :void calculateAverage() :double displayFeedback() :void
Admin	adminId :int adminName :String adminMail :String	displayAdmin() : void
FeedbackSystem	ArrayList<Student> students = new ArrayList< >(); ArrayList<Faculty> faculties = new ArrayList< >(); ArrayList<Course> courses = new ArrayList< >(); ArrayList<Question> questions = new ArrayList< >();	addStudent(Student s) :void addFaculty(Faculty f) :void addCourse(Course c) :void addQuestion(Question q) :void addFeedback(Feedback f) :void showAllStudents() :void showAllCourses() :void showAllQuestions() :void showAllFeedbacks() :void

	ArrayList<Feedback> feedbacks = new ArrayList< >();	generateCourseReport(int courseId)
File Manager	fileName :String	writeFeedbackToFile(ArrayList<Feedback> feedbacks) readFeedbackFromFile() appendFeedback(Feedback feedback) displayFileDetails()

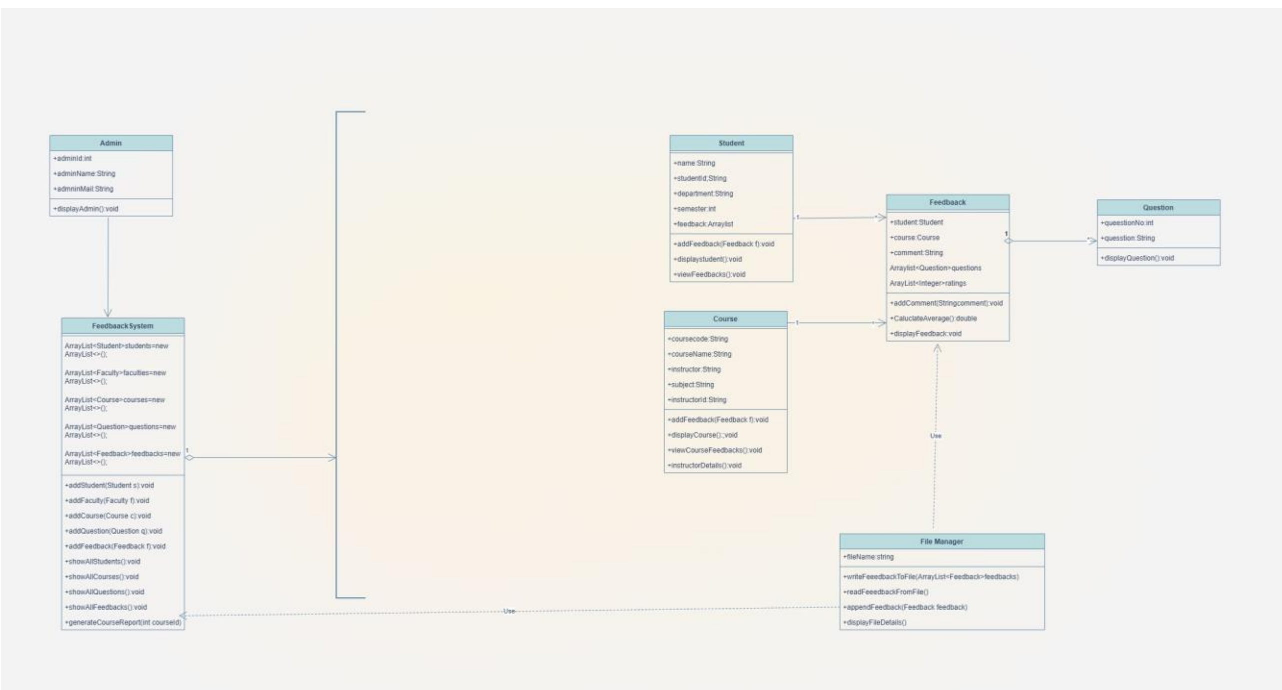
PHASE – II (Architectural Modelling)

UML Class Diagram and Sequence Diagram (CO2)

1. Problem Title : _________________ Course Feedback System _______________

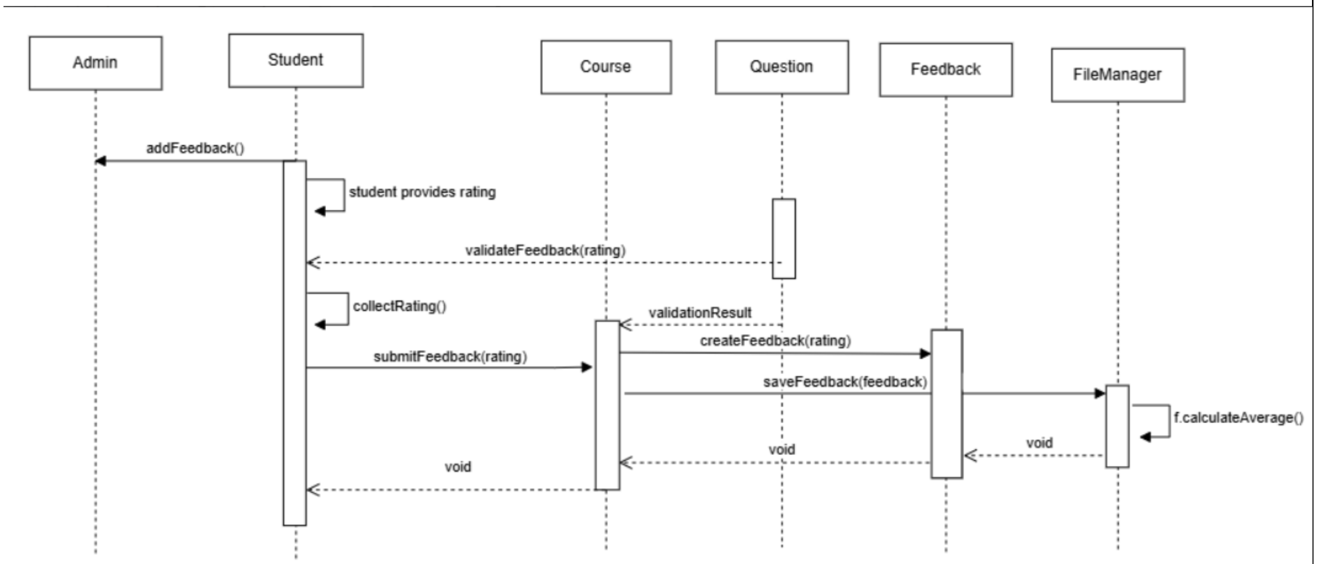
2. UML Class Diagram

Paste / Draw the UML Class Diagram here



3. Sequence Diagram

Paste / Draw the Sequence Diagram here



[seqdiagram.drawio]

4. Brief Explanation of the Design

- Major classes identified and their responsibilities:

1.Student : Stores student details, student gives feedback and an array contains feedback filled by students.
2.Course : Stores course and instructor/faculty information, holds feedbacks given by students.
3.Question : Stores feedback questions and collect responses from students
4.Feedback : Stores complete feedback given by a student, calculates average rating.
5.Admin : Stores admin information.
6.FeedbackSystem : It acts as the central controller that stores and manages arraylists and classes.
7.FileManager : Handles file operations.

- Relationships used (association / aggregation / inheritance, if any):
  - Association:

Student → Feedback(1 to many)
Course → Feedback(1 to many)
Admin → FeedbackSystem
FileManager→Feedback

  - Aggregation:

Feedback→Question
FeedbackSystem → All

- Depenndency

  FileManager - - - - -> Feedback

  - FileManager - - - - -> FeedbackSystem

- Key interaction flow captured in the sequence diagram:

The Admin starts the process and Student provides required details and rating. The Question class provides the criteria to ensure the rating is valid for that specific course. The Course creates a Feedback object, which performs the internal calculations. The FileManager takes the processed data and writes it to the physical File for permanent saving. The FeedbackSystem role is to show exactly which component handles logic, which handles data, and which handles storage.